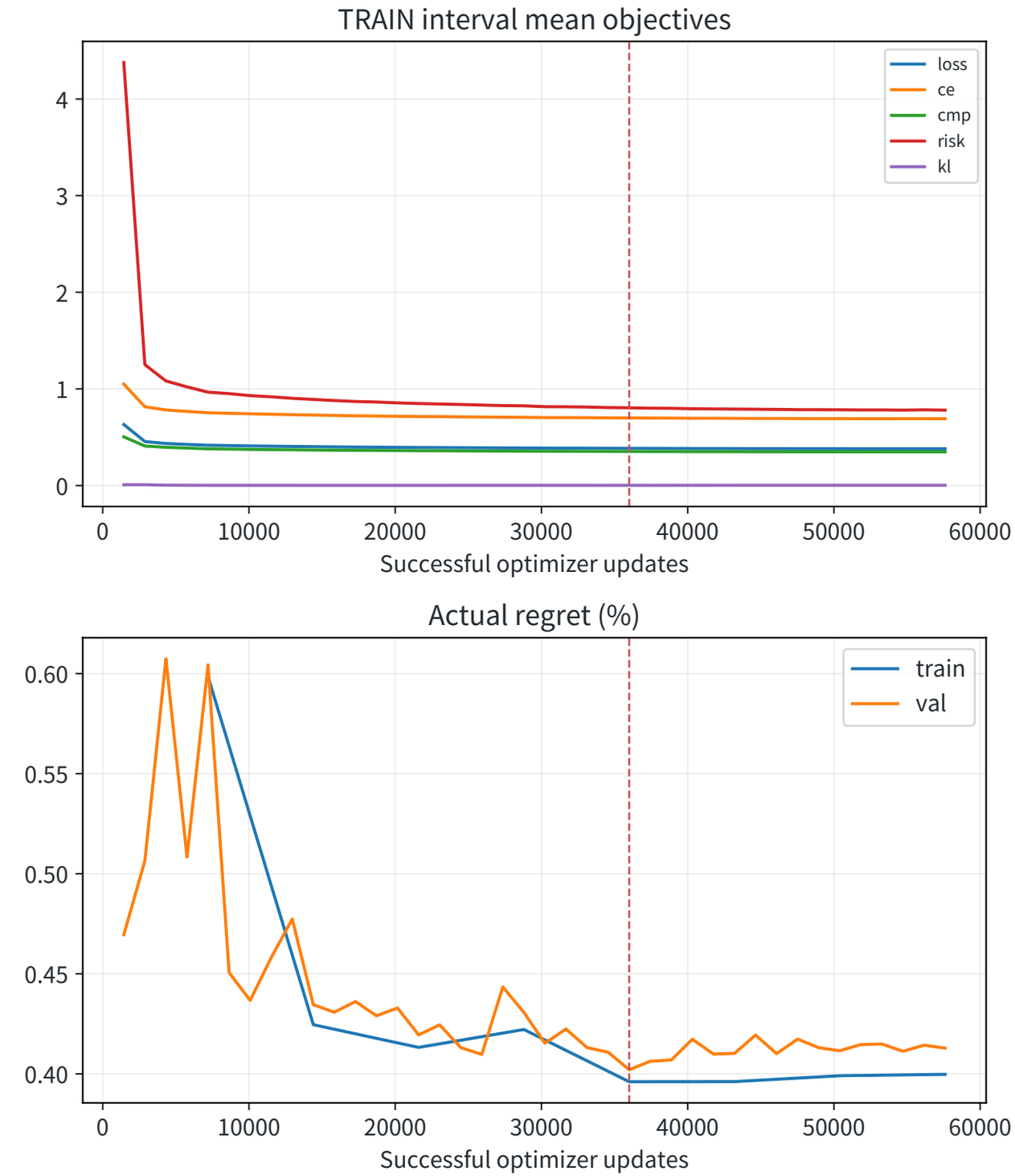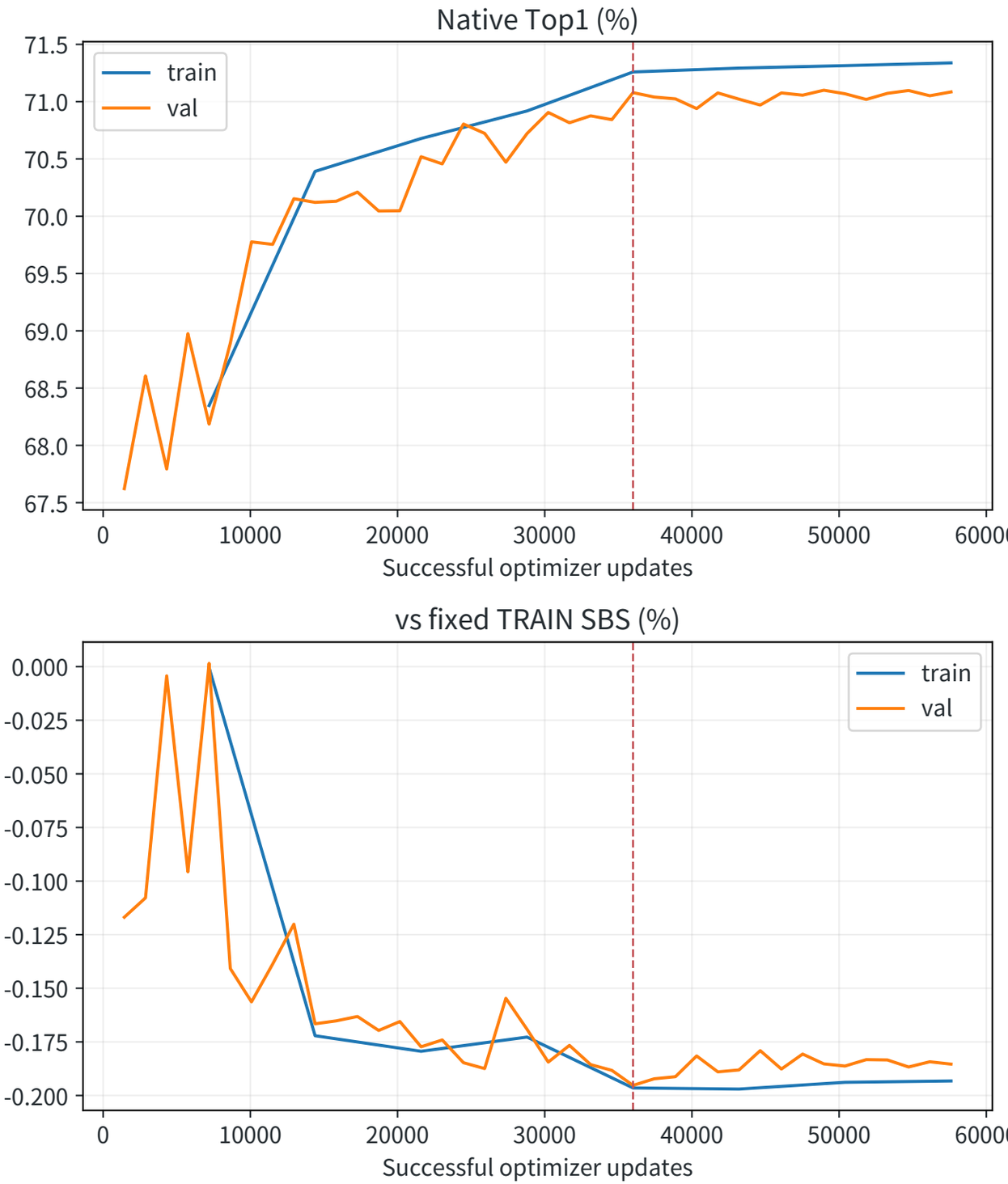R87/B_private_residual | successful updates; one locked VAL checkpoint